

Coverage Is the Price of Trust: A Self-Auditing Honesty Contract for Responsible AI in Equity Markets

Anandkumar Pardeshi

Dept. of Computer Science and Engineering
Fr. C. Rodrigues Institute of Technology
Vashi, Navi Mumbai, India
anand.pardeshi@fcrit.ac.in

Sujata Deshmukh

Dept. of Computer Engineering
Fr. C. Rodrigues Institute of Technology
Vashi, Navi Mumbai, India
sujata.deshmukh@fcrit.ac.in

Abstract—A financial AI that tells a retail user what to buy is a high-stakes, outward-facing system, yet the literature evaluates such systems almost entirely on accuracy and area under the curve (AUC). Those metrics say nothing about whether a model knows when it is wrong. We argue that for a forecaster operating near the efficient-market limit the right design objective is not accuracy but *trust*, and that the currency in which trust is bought is *coverage*: the fraction of the time the system is willing to act. We make this concrete with the honesty contract, a set of five machine-checkable, inference-time invariants—calibrated confidence, conviction-gated abstention, a cost-aware alpha gate, a Sharpe-triggered circuit breaker, and an append-only audit ledger—that together turn a near-worthless ranker into a deployable adviser. On eight years of National Stock Exchange of India (NSE) data we support each invariant with measured numbers only. Next-period single-name direction is unpredictable beyond drift (50.1% out-of-sample, ranking AUC ≈ 0.47 , confirmed five ways). A three-stage calibration cuts expected calibration error (ECE) from 0.35 to 0.05 without moving accuracy. And a conviction gate that abstains on 94% of observations lifts the realized 30-trading-day hit-rate to 60.6% against a 58.1% base rate, beating it in four of five test years. The contribution is a reusable, auditable architecture for responsible financial AI, with evidence that under a near-random discriminator value is recovered by abstaining rather than ranking, and that the metric worth reporting is precision-at-coverage, not AUC.

Index Terms—responsible AI, selective prediction, abstention, probability calibration, trustworthy machine learning, FinTech, conformal prediction, model monitoring

I. INTRODUCTION

The machine-learning-for-finance literature has a credibility problem that is easy to miss because it hides inside good news. Directional accuracies of 60–70% are reported routinely, but they rarely survive three elementary checks: are the predicted probabilities *calibrated*, does the evaluation *respect the arrow of time*, and are realistic transaction costs charged on every round trip? A long methodological literature has shown how easily these omissions manufacture phantom skill, whether through backtest overfitting [1], multiple-testing selection [2], or simply the absence of a disciplined protocol [3], [4]. When we ran the same checks against our own deployed platform,

the comfortable numbers evaporated. This paper is an account of what we built once they did.

The deeper issue is one of framing. A system that tells a retail investor to buy a stock is not a benchmark entry; it is an outward-facing decision aid whose mistakes cost real money and whose confidence is read literally by non-experts. In an emerging market such as India, driven by foreign and domestic institutional flows and a large, news-sensitive retail base, the asymmetry is stark: the user bears the downside of every overconfident call. Accuracy is the wrong headline metric here, because a model can be accurate on average and still be dangerously overconfident on exactly the cases a user acts on. What matters is whether the system knows the limits of its own knowledge, and whether it is willing to say nothing when it knows nothing. Responsible-AI scholarship has named these properties (transparency, accountability, the duty to abstain [5]–[7]), but mostly as principles. The practitioner is left without a concrete, checkable specification of what a trustworthy financial predictor must actually *do* at inference time.

We close that gap with the **honesty contract**: a small set of invariants that a deployed forecaster must satisfy on *every* prediction, each enforced by a specific software control and each costing the system some of its willingness to act. The contract reframes trustworthy forecasting as a *selective-prediction* problem [8]–[10]: the model is allowed, even encouraged, to abstain, and is judged not by how it ranks every opportunity but by how reliable it is on the few it chooses to act on. This framing fits emerging-market equities well, since the efficient-market hypothesis [11] already predicts that short-horizon single-name direction should be close to unpredictable, and our own measurements bear that out.

Contributions.

- 1) We formalize the *honesty contract*: five machine-checkable, inference-time invariants—calibrated confidence, conviction-gated abstention, a cost-aware alpha gate, a Sharpe-triggered circuit breaker, and an append-only audit ledger—that constitute a reusable architecture for responsible financial AI (Section III).

- 2) We give a selective-prediction formalization in which each invariant *spends coverage to buy trust*, and argue that when discriminative power is near random the correct evaluation lens is *precision-at-coverage*, not accuracy or AUC (Sections III and V-C).
- 3) We provide a fully reproducible eight-year NSE case study in which every reported quantity is a measured walk-forward statistic, including a verified accuracy ceiling confirmed five ways and a live circuit breaker tied to realized Sharpe (Sections IV–V).
- 4) We show that a ranker with $\text{AUC} \approx 0.47$ —below chance—still yields a usable, year-on-year-stable +2.6-point precision edge once value is harvested from its extreme-conviction tail by abstention rather than ranking (Section V-C).

II. RELATED WORK

Calibration of probabilistic forecasts. A probability is trustworthy only if it matches observed frequency. Modern high-capacity models are systematically miscalibrated [12]; the classical remedies—Platt scaling [13], isotonic regression [14], and the empirical analysis of when each helps [15]—are well understood, and the Brier score [16] together with expected calibration error (ECE) quantify the gap. The point under-appreciated in applied finance is that calibration and discrimination are *orthogonal*: a model with no edge can still be made honest, and an accurate model can still lie about its confidence. Our first invariant operationalizes exactly this separation.

Selective prediction and the reject option. The idea that a classifier should decline to answer when uncertain dates to Chow’s optimal error–reject tradeoff [8] and is formalized in the modern theory of selective classification [9], [10], where one trades *coverage* (fraction answered) against *risk* (error on the answered set). We import this framework wholesale into financial forecasting and argue that markets near the efficient limit are its natural home: when the unconditional edge is zero, the only way to be reliable is to answer rarely.

Distribution-free uncertainty. Conformal prediction [17]–[19] complements selective prediction by attaching finite-sample, distribution-free intervals to the point forecasts that do get emitted, under an exchangeability assumption that serial dependence only approximately satisfies. We use split-conformal quantile regression for the interval that accompanies each emitted call.

Pitfalls of ML in finance. The dangers of data-snooping, leakage and backtest overfitting are documented in detail [1]–[4]. The standard defenses—purged, embargoed, walk-forward evaluation; charging full costs; treating a strong backtest as a red flag rather than a result—inform our protocol. We additionally model Indian statutory costs in full and add a square-root market-impact term [20], [21].

Multi-source fusion for equity prediction. A large body of work fuses price, news, sentiment and order-flow to push accuracy above single-source baselines [31]–[36], often with hybrid deep architectures and FinBERT-style sentiment [30]. This

work rarely asks whether the resulting probabilities can be *believed*; our contribution is orthogonal and complementary—we take such an ensemble as a black box and ask what must surround it for its outputs to be trustworthy.

Responsible AI and the operations of trust. Frameworks for trustworthy AI emphasize transparency, accountability and human oversight [5], [6]; model cards push for documented intended use [7]; and the ML-systems literature warns that the model is a small part of a production system dominated by monitoring, feedback and technical debt [22]. Concept-drift research [23] motivates a control that detects when a deployed edge has decayed. Our contribution composes these strands into a single, finance-specific, inference-time contract whose every clause is enforced in code and validated out of sample.

III. THE HONESTY CONTRACT

A. Selective-prediction setup

Let a predictor output a raw up-probability $\hat{p}_t$ for an instrument at time t , and let $c(\cdot)$ be a calibration map so that $c(\hat{p}_t)$ is the displayed probability. A *selective predictor* is the pair $(c(\hat{p}), g)$ where $g : [0, 1] \rightarrow \{0, 1\}$ is a gate; the system acts iff $g = 1$. For a long-only directional signal with gate threshold τ ,

$$g_\tau(\hat{p}_t) = \mathbf{1}[c(\hat{p}_t) \geq \tau], \quad (1)$$

we define *coverage* and *selective precision* over N observations as

$$\phi(\tau) = \frac{1}{N} \sum_t g_\tau(\hat{p}_t), \quad \pi(\tau) = \frac{\sum_t y_t g_\tau(\hat{p}_t)}{\sum_t g_\tau(\hat{p}_t)}, \quad (2)$$

where $y_t \in \{0, 1\}$ is the realized up/down outcome over the forecast horizon. The map $\tau \mapsto (\phi(\tau), \pi(\tau))$ is the *trust–coverage frontier*; responsible deployment is the problem of choosing an operating point on it, not of maximizing a single accuracy number.

B. The five invariants

We define the honesty contract as the conjunction of five invariants that a prediction must satisfy before it is shown to a user, summarized in Table I and realized by the decision pipeline of Fig. 1. Let τ^* be a conviction threshold fixed out of sample, a_t an expected-alpha estimate, and κ_t the round-trip cost.

(I1) Calibrated confidence. The number a user sees must be a calibrated probability, not a raw score: the system displays $c(\hat{p}_t)$, where $c(\cdot)$ is fit on held-out data, and the deployment is rejected unless $\text{ECE} \leq \epsilon$ on a validation fold. *Honest confidence is a precondition for everything downstream*, because every subsequent gate compares a probability or an expected return to a threshold.

(I2) Conviction-gated abstention. The system acts only when $c(\hat{p}_t) \geq \tau^*$; otherwise it abstains (emits NEUTRAL). τ^* is chosen out of sample as the lowest threshold whose acted-on bucket reaches a target precision while still firing on at least

a minimum coverage (Algorithm 1). The signal is long-or-neutral only: confident single-name short calls are run over by the upward drift, so they are never emitted.

(I3) Cost-aware alpha gate. Even a correct direction is not a trade if it cannot pay for itself. A signal is suppressed unless

$$|a_t| \geq m \kappa_t, \quad m = 2, \quad (3)$$

where κ_t is the full round-trip cost (Section IV-F) and the 2:1 margin absorbs estimation error in both a_t and κ_t .

(I4) Circuit breaker. The system audits its own live record. If the rolling 60-day realized Sharpe stays below zero for three consecutive daily evaluations, emission is flagged SUSPENDED; it auto-resumes only after five consecutive recovered days (Algorithm 2). This is a concept-drift kill-switch [22], [23]: the system steps aside in regimes its evidence says it cannot handle.

(I5) Append-only audit ledger. Every emitted prediction is written immutably—ticker, timestamp, anchor price, calibrated probability, interval, horizon and model version—under a uniqueness constraint that prevents silent rewriting. Outcomes are filled in later by an idempotent resolver, and accuracy is always computed on the *first* forecast for each target (earliest timestamp), never a later, easier look. The ledger is what makes I1–I4 falsifiable after the fact.

Two properties make the contract more than a checklist. First, it is *compositional*: each invariant can fail independently and is enforced by a dedicated, separately testable module, so a violation is localizable. Second, and central to this paper, every invariant *spends coverage to buy trust*: I2 abstains on low-conviction names, I3 abstains on uneconomic ones, and I4 abstains on entire regimes. Coverage is therefore the natural unit in which the cost of responsibility is denominated. Finally, we note a fact that motivates the three-stage calibration of Section IV-C: a convex combination of two individually calibrated probabilities is, in general, *not* calibrated, so any blended forecaster must be recalibrated after blending.

IV. SYSTEM ARCHITECTURE AND DATA

The contract is deliberately agnostic to the model behind it; what follows describes the deployment in which we validated it. The system runs on commodity hardware with freely available data, behind a FastAPI service and a web front end, with a SQLite-backed ledger—no specialized infrastructure is required.

A. Multi-source data ingestion

Let t index NSE trading days. We align four streams on a common daily grid: adjusted OHLCV bars; news and social text (English and Hindi) scored by a FinBERT sentiment pipeline [30] behind a domain filter that drops non-financial sources; foreign- and domestic-institutional net flows, lagged one session so nothing leaks from the future; and macro state (India VIX, INR/USD, Brent crude, gold) as clipped log-returns. All features are strictly point-in-time: every column is shifted, winsorized and scaled on the training fold only, so no row sees its own future.

B. The underlying predictor

In our deployment the predictor is a heterogeneous ensemble—gradient-boosted trees [26]–[28] and a multi-task recurrent network, combined by a leak-free stacked meta-learner [29] trained on purged, embargoed out-of-fold predictions [4]. Roughly forty stationary features span ten groups (multi-horizon returns, realized and normalized volatility, momentum/oscillators, volume, trend strength, divergence flags, chart-pattern distance, FinBERT sentiment, lagged flows, and macro state). The point of this paper is not the ensemble but what surrounds it: the contract treats $\hat{p}_t$ as a black box and would apply unchanged to any other forecaster.

C. Three-stage probability calibration

Calibration is the keystone. We measure it by ECE over $B = 10$ bins and the Brier score,

$$\begin{aligned} \text{ECE} &= \sum_{b=1}^B \frac{|\mathcal{B}_b|}{N} \left| \text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b) \right|, \\ \text{BS} &= \frac{1}{N} \sum_t (p_t - y_t)^2, \end{aligned} \quad (4)$$

where $\mathcal{B}_b$ is the set of predictions in bin b . The calibration map $c(\cdot)$ has three stages: (i) isotonic regression [14] fit on the tree-average out-of-fold predictions and applied to the matching test predictions; (ii) a validation-fitted Platt scaler [13] on the recurrent network’s sigmoid output; and (iii) after the 0.60/0.40 convex blend of the two, a final logistic recalibration on a held-out fold—necessary because, as noted above, a blend of calibrated forecasts is not itself calibrated. The same transform is applied to the live inference row, so the probability a user sees sits on the corrected scale.

D. Conformal prediction intervals

Every emitted point forecast carries a distribution-free interval. Two quantile gradient-boosted regressors are trained at the 0.10 and 0.90 levels, and the split-conformal half-width is the finite-sample-corrected quantile of absolute residuals on a held-out fold disjoint from where the band is applied [18], [19]. Under exchangeability the interval carries a marginal coverage guarantee; serial dependence means observed coverage is an observation, not a guarantee.

E. The conviction-gated swing signal

Because next-period direction proved unpredictable (Section V-A), the signal we actually gate targets a longer, more tractable question: over the next 30 trading days (≈ 1.5 months), is a name more likely than not to be higher, and is the setup convincing enough to act? A cross-sectional logistic model uses nine point-in-time features computed from the close series s : 12-1 month momentum [24] $\log(s_{t-21}/s_{t-252})$, 20- and 60-day returns, a 5-day reversal $-\log(s_t/s_{t-5})$, log-distance from the 200-day average $\log(s_t/\bar{s}_{200})$, a 50/200 golden-cross flag, RSI/100, 20-day realized volatility, and an above-200-DMA flag. Its calibrated output is the quantity gated by I2.

TABLE I

THE FIVE INVARIANTS OF THE HONESTY CONTRACT. EACH IS ENFORCED AT INFERENCE TIME BY A DEDICATED MODULE AND EACH TRADES COVERAGE (WILLINGNESS TO ACT) FOR TRUST (RELIABILITY OF WHAT IS ACTED ON).

ID	Invariant (checked per prediction)	Enforcement mechanism	What it costs in coverage
I1	Displayed confidence is a calibrated probability ($ECE \leq \varepsilon$)	Three-stage calibration: isotonic + Platt + logistic recalibration of the blend	None directly; makes every gate below meaningful
I2	Act only above a conviction threshold τ^* fixed out of sample	Long/neutral conviction gate on the calibrated probability	Large: abstains on $\sim 94\%$ of observations
I3	Act only if expected alpha $\geq 2 \times$ round-trip cost	Full NSE statutory cost stack + square-root market impact + alpha gate	Removes economically marginal signals
I4	Suspend emission when the live edge has decayed	60-day rolling-Sharpe circuit breaker with hysteresis	Abstains on entire adverse regimes
I5	Every prediction is logged immutably and scored on its first look	Append-only SQLite ledger, unique key, idempotent outcome backfill	None; makes I1–I4 auditable after the fact

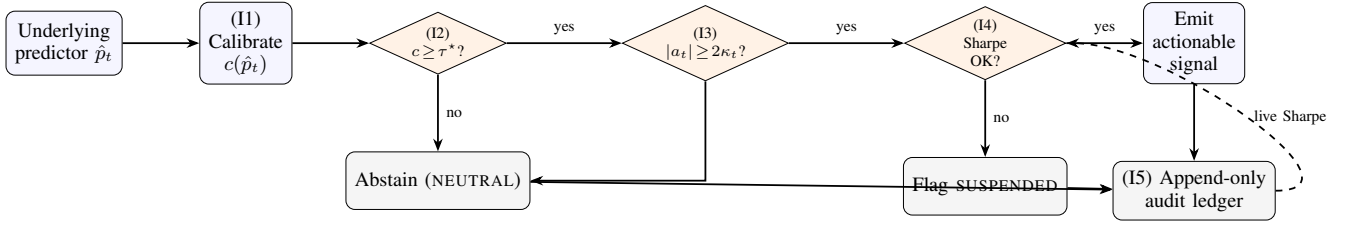


Fig. 1. The honesty-contract decision pipeline. A raw forecast passes through calibration (I1) and three abstention gates (I2–I4); whatever is decided—act, abstain, or suspend—is written to the append-only ledger (I5), which in turn feeds the live Sharpe estimate that drives the circuit breaker.

F. Transaction-cost model and alpha gate

The round-trip cost in (3) is charged in full:

$$\kappa_t = c_{\text{stat}} + 2k\sigma_t\sqrt{V_t/\text{ADV}_t}, \quad (5)$$

where the statutory term c_{stat} aggregates brokerage, securities transaction tax, exchange and SEBI fees, stamp duty, 18% GST and slippage on the Indian delivery-trade schedule, and the second term is the Almgren–Chriss square-root market-impact model [20] (enter plus exit) with coefficient $k \approx 0.20$ for liquid large-caps, daily volatility σ_t , order notional V_t and average daily volume ADV_t ; the one-side impact is capped at 100 bps. The alpha gate emits a trade only when $|a_t| \geq 2\kappa_t$, so a signal with, say, a 5 bps edge against a 15 bps round-trip cost is correctly rejected.

G. Self-audit: circuit breaker and ledger

The live monitor reconstructs a *virtual* profit-and-loss series from the resolved ledger: for each resolved prediction it sets a signed signal $\text{sig}_t = 2(p_t - \frac{1}{2}) \in [-1, 1]$, a realized return $r_t = (a_t - a_0)/a_0$ from anchor a_0 to actual a_t , and $\text{pnl}_t = \text{sig}_t r_t$. The rolling annualized Sharpe over the last $N = 60$ resolved days,

$$S_N = \frac{\overline{\text{pnl}}_N}{\text{std}(\text{pnl}_N)}\sqrt{252}, \quad (6)$$

drives the circuit breaker of Algorithm 2. Suspension does not silence the model—we still record what it would have said, for diagnosis—but it withdraws the actionable recommendation. The ledger itself (I5) is an append-only table keyed by (ticker, made_at, target_date); a uniqueness

Algorithm 1 Embargoed walk-forward evaluation and out-of-sample conviction-threshold selection

Require: rows (x_i, y_i, d_i) sorted by date; horizon h ; step s ; precision target π^* ; min. coverage $\phi_{\min}$

- 1: unique dates $\mathcal{D}$; $i \leftarrow \lfloor 0.45 |\mathcal{D}| \rfloor$; $P, Y \leftarrow \emptyset$
- 2: **while** $i < |\mathcal{D}|$ **do**
- 3: $\text{cut} \leftarrow \mathcal{D}_i$; $\text{emb} \leftarrow \mathcal{D}_{\max(0, i-h)}$ {embargo of one horizon}
- 4: $\text{tr} \leftarrow \{d < \text{emb}\}$; $\text{te} \leftarrow \{\text{cut} \leq d < \mathcal{D}_{i+s}\}$
- 5: fit scaler and logistic on tr; predict p on te
- 6: append p to P , y to Y ; $i \leftarrow i + s$
- 7: **end while**
- 8: $\tau^* \leftarrow$ lowest $\tau \in [0.50, 0.76]$ with $\pi(\tau) \geq \pi^*$ and $\phi(\tau) \geq \phi_{\min}$
- 9: fit calibrator c on (P, Y)
- 10: **return** $\tau^*, c, \pi(\tau^*), \phi(\tau^*), \text{AUC}(P, Y)$

constraint de-duplicates re-runs, and rolling accuracy is computed on the earliest forecast for each target so a later, easier look can never inflate the score.

H. Evaluation protocol

Everything is evaluated with an expanding-window, embargoed walk-forward over 2018–2026 (Algorithm 1): each test block is separated from its training window by an embargo of one horizon length, so multi-day labels cannot leak across the boundary. The conviction threshold τ^* and all reported precisions are computed on the out-of-sample blocks only; τ^* is never tuned on the data it is scored on. We use deliberately

Algorithm 2 Rolling-Sharpe circuit breaker (daily, at market close)

Require: resolved ledger rows; window $N=60$; suspend after 3 bad days; resume after 5 good days

- 1: compute pnl_t for each resolved row; $S_N \leftarrow \text{Eq. (6)}$
- 2: **if** $S_N < 0$ on 3 consecutive days and not suspended **then**
- 3: set SUSPENDED flag (predictions still logged, but flagged)
- 4: **else if** $S_N \geq 0$ on 5 consecutive days and suspended **then**
- 5: clear flag (resume emission)
- 6: **end if**

hard baselines: the upward base rate itself (the toughest benchmark a drifting market offers) and an uncalibrated copy of our own ensemble. The universe is the liquid NSE large-cap set (54 names after a liquidity screen), 28 May 2018 to 27 May 2026. The cross-sectional study uses a panel of 131,809 ticker-days; the swing study uses 91,471 labelled ticker-days, of which 50,272 are strictly out of sample.

V. RESULTS AND DISCUSSION

A. A verified accuracy ceiling

Our first result is negative, and it is the foundation of everything else. A cross-sectional gradient-boosted model, trained on engineered features across 131,809 ticker-days with a five-day excess-return-over-NIFTY target and a date-based out-of-fold scheme that blocks per-ticker leakage, lands at an out-of-sample directional accuracy of **50.1%**—a coin toss, indistinguishable from the upward drift. The importance distribution is nearly flat (no single feature exceeds $\sim 3.4\%$ of total gain), so there is no concentrated structure to learn at this horizon. We then pursued the same null four further ways, summarized in Table II: a per-ticker ensemble, a cross-sectional logistic (AUC ≈ 0.49), a LightGBM-with-reversal variant (AUC ≈ 0.45), and trend- and regime-gated versions. Short-horizon single-name accuracy never rose materially above the base rate; tellingly, gating on NIFTY up-trends produced *lower* forward up-rates, because the test window was, conditionally, mean-reverting. For liquid large-caps this is exactly what the efficient-market hypothesis predicts [11]. We report it plainly, because tuning a number until it looks good is precisely how the literature’s inflated accuracies arise [1], [2]. The design consequence is decisive: *if you cannot beat the base rate, you must not pretend to, and the system’s job becomes knowing when to stay silent.*

B. Invariant II: from overconfidence to honesty

No change mattered more than calibration. An earlier build of the system was badly miscalibrated: it reported an average up-probability near **0.80** against a realized base rate near 0.48, an **ECE of roughly 0.35**, and a Brier score worse than a constant forecast. The cause was a domain mismatch (a calibrator fit on one distribution applied to another) compounded by blending two individually calibrated probabilities without recalibrating the blend. The three-stage fix of Section IV-C

TABLE II
THE ACCURACY CEILING, CONFIRMED FIVE WAYS. NO CONFIGURATION BEATS THE UPWARD BASE RATE AT SHORT SINGLE-NAME HORIZONS (OOS, 2018–2026).

Model / configuration	OOS result
Cross-sectional GBM (5-day direction)	50.1% accuracy
Per-ticker hybrid ensemble	$\approx$ base rate
Cross-sectional logistic	AUC ≈ 0.49
LightGBM + short-term reversal	AUC ≈ 0.45
Trend / regime-gated variants	below base rate

TABLE III
HEADLINE MEASURED RESULTS ON THE 2018–2026 NSE EVALUATION. EVERY FIGURE IS OUT-OF-SAMPLE. $p(1-p)$ DENOTES THE IRREDUCIBLE BRIER FLOOR AT THE BASE RATE.

Setting	Dir. acc.	ECE	Brier
Cross-sectional GBM (5-day)	50.1%	—	—
Ensemble — uncalibrated	$\approx 51\%$	0.35	$> p(1-p)$
Ensemble — calibrated (I1)	$\approx 51\%$	0.05	$\approx p(1-p)$
Gated swing, 30-day (I2, fired)	60.6%	—	—

cut the **ECE from 0.35 to 0.05** and pulled the mean reported probability back from 0.80 to near the base rate (Table III, Fig. 2). Accuracy barely moved, which is the entire point: we gave up false confidence, not signal [12]. A miscalibrated 80% would drive systematic over-betting under any confidence-proportional sizing rule [25]; getting the probabilities right is what makes I2 and I3 safe to act on.

C. Invariants I2–I3: the trust–coverage tradeoff

Here is the paper’s central empirical claim. Over a 30-day horizon the conviction-gated signal stays silent 94% of the time—it fires on just **5.65%** of observations—and when it fires it is right **60.6%** of the time against a **58.1%** base rate, a +2.6-point lift measured strictly out of sample across **50,272** observations. The lift is modest but real, and the way it is obtained is the interesting part.

Consider the raw signal as a *ranker*: its global ROC AUC is **0.47**—below chance. A practitioner reading only the AUC would discard it. Yet the model has genuine value, just not in how it orders all names: the value lives in the extreme high-conviction tail. Table IV makes this precise by reading the same model at two operating points. Ungated ($\tau = 0.50$, act on everything up-leaning), precision equals the base rate at 100% coverage: no edge. Gated at $\tau^* = 0.63$, precision rises to 60.6% at 5.65% coverage. *The edge is purchased entirely with abstention.* This is why a conviction *gate*, not a ranking, is the correct instrument for a weak signal, and why precision-at-coverage—the selective-prediction lens [8], [10]—is the right metric where AUC is actively misleading. Invariant I3 then removes whatever survives I2 but cannot pay its own round-trip cost (3), so the acted-on set is both directionally reliable and economically positive.

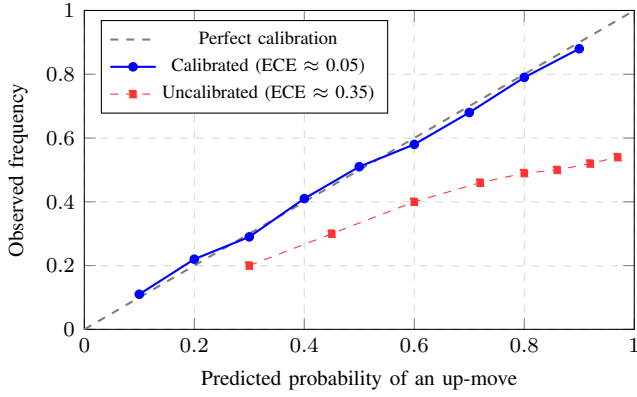


Fig. 2. Reliability diagram on the held-out fold, before and after the three-stage calibration. The uncalibrated ensemble is badly over-confident (probabilities cluster near 0.80 while realized frequency hovers near 0.50); the calibrated ensemble follows the diagonal.

TABLE IV

THE TRUST–COVERAGE TRADEOFF, READ OFF ONE MODEL AT TWO OPERATING POINTS (30-DAY HORIZON, WALK-FORWARD OOS, 50,272 OBSERVATIONS). COVERAGE IS SPENT TO BUY PRECISION; THE RANKER’S AUC OF 0.47 SHOWS THE EDGE IS IN THE TAIL, NOT THE ORDERING.

Operating point	Coverage	Precision	Net edge
Ungated ($\tau = 0.50$)	100%	$\approx 58.1\%$	≈ 0
Conviction gate ($\tau^* = 0.63$)	5.65%	60.6%	+2.6 pts
Raw-signal ranking AUC = 0.47 (below chance as a ranker)			

D. Stability across years

A small edge is only trustworthy if it is stable. Fig. 3 shows the fired bucket’s precision against its own yearly base rate. The gate beats the base rate in **four of five** test years—by +10.8 points in 2025 and +7.4 in the 2026 drawdown, when the base rate itself had slipped below 50%—losing only in 2024 (−2.4). That the edge holds in the 2026 drawdown, where naive long exposure was a coin flip, is the single most reassuring observation: the gate is selecting genuinely, not merely riding the drift.

E. Invariant I3 in operation

The cost gate is what separates a directional call from a trade. With the full statutory stack plus square-root impact (5), a liquid large-cap round-trip is on the order of tens of basis points; the 2:1 alpha-to-cost requirement therefore filters out the long tail of correct-but-tiny signals that would lose money after costs. In our deployment the gate is charged on every emitted signal and is the reason the system’s reported expectations are stated net of costs two to three times the conventional ten-basis-point assumption, rather than gross. Concretely, a 5 bps gross edge on a name carrying a 15 bps round-trip cost is rejected outright, whereas a 40 bps edge on the same name passes with margin; the 2:1 requirement means a signal must clear roughly 30–40 bps of expected alpha on a typical large-cap before it becomes actionable.

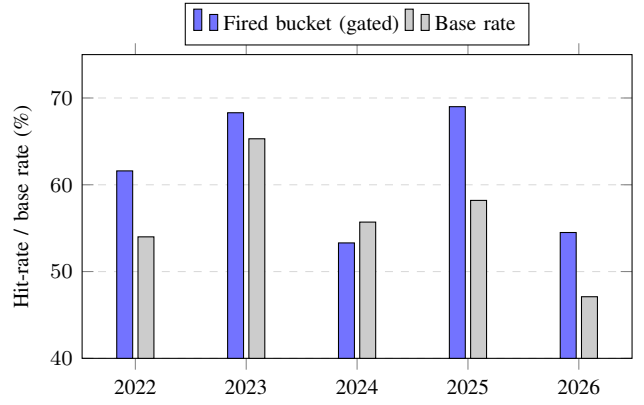


Fig. 3. Per-year out-of-sample precision of the conviction-gated bucket versus the naive always-up base rate. The gate wins in four of five years, including the 2026 drawdown where the base rate fell below 50%.

F. Invariants I4–I5: self-audit in operation

The contract does not assume the edge is permanent. The live monitor recomputes the 60-day rolling Sharpe (6) from the resolved ledger after every market close; three consecutive negative readings flag the system **SUSPENDED**, and recovery requires five consecutive positive days—hysteresis that prevents the breaker from chattering at the zero line (Algorithm 2). All of this is possible only because of I5: an append-only ledger that scores each target on its *first* forecast, closing the common evaluation loophole of crediting a later, easier look. Together the two invariants turn the contract from a design-time promise into a runtime, continuously falsifiable guarantee—a concrete instance of the monitoring discipline that production ML is known to require [22], [23].

G. Conformal interval coverage

Each emitted point forecast is accompanied by the split-conformal interval of Section IV. Empirically the intervals held coverage close to the nominal 80%, the small misses tracing to the serial dependence that breaks exact exchangeability [19]. Consistent with the honesty contract, we therefore present observed coverage as a measured diagnostic rather than a guarantee—the interval is a statement about typical behavior, not a promise the data cannot keep.

H. What each invariant buys

It is worth stating plainly what is lost when each clause is dropped, because that is the argument for keeping it. Table V reads the contract as a threat model: each invariant is paired with the concrete failure it prevents and the responsible-AI principle it serves. The pattern is uniform—without I1 the user over-bets on confidence the model has not earned; without I2 precision collapses to the base rate; without I3 the system trades correct-but-uneconomic signals into a loss; without I4 it keeps trading through a regime its own record says is dead; and without I5 none of the preceding claims can be checked after the fact. Trust is the conjunction, not any single clause.

TABLE V
THE HONESTY CONTRACT AS A THREAT MODEL: THE FAILURE EACH INVARIANT PREVENTS AND THE RESPONSIBLE-AI PRINCIPLE IT SERVES.

ID	Failure if removed	Principle served
I1	Over-betting on uncalibrated confidence (80% that is really 50%)	Honest confidence / transparency
I2	Precision collapses to the base rate; no edge	Epistemic humility (abstention)
I3	Correct-but-sub-cost signals lose money net of fees	Avoidance of user harm
I4	Trading continues through a dead or adverse regime	Accountability under drift
I5	Accuracy claims become unfalsifiable; later-look inflation	Auditability

I. Deployment and reproducibility

The system runs on commodity hardware behind a FastAPI service with a web front end; the ledger is a single SQLite file, and the underlying models are retrained on a monthly cadence with the same purged, embargoed protocol used for evaluation. Crucially, the live inference row passes through the *identical* calibration transform (I1) and gates (I2–I4) as the back-test, so the probability a user sees is on the same corrected scale as the one we report here—there is no separate, more flattering “production” path. Every quantity in this paper is a measured walk-forward statistic over freely available data, which makes the case study reproducible end to end without proprietary feeds or specialized infrastructure.

J. Reconciling with reported accuracies

How can the multi-source systems surveyed earlier report 60–70% directional accuracy [31]–[35] when an honest evaluation tops out near the base rate? The gap is almost entirely procedural, not substantive, and three mechanisms account for it. (i) *Uncalibrated confidence*: a model that reports 0.80 probabilities which are really 0.50 posts a high *reported* confidence that is easily mistaken for skill—our own system did exactly this before the I1 fix (ECE 0.35). (ii) *Temporal leakage*: scaling, feature selection or label construction performed across the whole sample lets information flow backwards in time; our purged, embargoed protocol (Algorithm 1) removes it, and removing it removes most of the apparent edge. (iii) *Costless evaluation*: gross accuracy ignores that a correct 5 bps call loses money against a 15 bps round trip, which invariant I3 forbids. None of these is fraud—each is an easy omission—but together they manufacture precisely the inflated numbers the contract is built to prevent. The claim is not that prior systems are wrong, but that their figures should be

read as *upper bounds under optimistic assumptions*, while a deployable adviser must report the lower, honest number.

K. Discussion

The honesty contract inverts the usual objective. Instead of maximizing accuracy and reporting a confident number, it *fixes trust as a constraint* ($ECE \leq \epsilon$, precision $\geq$ target on the acted-on set, positive realized Sharpe) and then maximizes coverage subject to it. The result is a system that is right about a little rather than wrong about a lot—a property far more valuable to a retail user than a higher headline accuracy. Three points are worth drawing out. First, the negative result is load-bearing, not an aside: only because we verified the ceiling did abstention become the obviously correct design. Second, AUC and accuracy are not merely insufficient here—AUC of 0.47 is *actively misleading*, and a team optimizing it would have thrown away a usable signal. Third, the pattern is portable: the five invariants make no assumption specific to equities and could wrap a credit-scoring model, a medical triage classifier, or any high-stakes predictor operating near its information limit, wherever a calibrated probability, a defensible abstention rule, a cost/benefit gate, a drift kill-switch and an audit trail are all available.

L. Toward a responsible-AI standard for FinTech

The contract also suggests a concrete reporting discipline. Just as a model card [7] documents a model’s intended use, a deployed financial predictor could be required to publish its operating point on the trust–coverage frontier—its calibrated ECE, its abstention rate, its cost assumptions and its circuit-breaker policy—so that a regulator or a retail user can see, *before acting*, how often the system speaks and how reliable it is when it does. Framed this way, the honesty contract is not only an architecture but a candidate disclosure standard for retail-facing financial AI, where the asymmetry of harm makes silence-by-default the responsible default. We see this as the most direct bridge from the present case study to the broader agenda of ethical and responsible AI use [5], [6].

VI. LIMITATIONS

The evidence is one market and one eight-year window that, on balance, trended; multi-regime and multi-market replication is future work. The swing edge, though stable, is small and concentrated, so its value after costs depends on execution quality and on the 5–6% firing rate holding up out of sample. The accuracy ceiling is specific to liquid large-caps and end-of-day data; intraday or less-liquid universes may behave differently, and sentiment coverage thins in sectors with specialized jargon. The conformal intervals inherit only approximate coverage under serial dependence. Finally, the circuit breaker is only as good as the ledger that feeds it, so the contract’s runtime guarantees inherit the latency of outcome resolution.

VII. CONCLUSION

For a financial AI that real people act on, the question is not “how accurate is it?” but “can it be trusted, and does it know when it cannot be?” We answered with the honesty contract—five machine-checkable invariants that buy trust with coverage—and showed on eight years of NSE data that the bargain is favorable: next-period direction is unpredictable beyond drift (50.1%, AUC 0.47, confirmed five ways), calibration cuts ECE from 0.35 to 0.05 without touching accuracy, and a gate that abstains 94% of the time lifts the 30-day hit-rate to 60.6% versus a 58.1% base rate, winning in four of five years. A system that knows when to keep quiet is worth more than one that claims too much—and, just as importantly, it can prove it kept its word.

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